*Inspired by the RFM chapter in the book **R All-in-One For Dummies** written by Joseph Schmuller*

Background RFM analysis stands for Recency-Frequency-Monetary values. The UC Irvine Machine Learning Repository has the Online Retail dataset - which contains all the transactions occurring between 01/12/2010 and 09/12/2011 for a UK-based and registered non-store online retail.

Data Columns : InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country

Goal

1. We want to identify the top customers based on RFM results. Who are the top 20%?
2. Based on the RFM scores, identify 'Champion' and 'Lost' customers.
3. Summarize the key difference between the two customer segments.

ChatGPT

```
install.packages('readxl')
library(readxl)
```

```
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)
```

```
retailonline <- read_xlsx('/OnlineRetail.xlsx')
head(retailonline)
```

A tibble: 6 × 8

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
<chr>	<chr>	<chr>	<dbl>	<dttm>	<dbl>	<dbl>	<chr>
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 08:26:00	7.65	17850	United Kingdom

```
#RFM in R
install.packages('didrooRFM')
library(didrooRFM)
retailonline$Amount <- retailonline$Quantity*retailonline$UnitPrice

#Prepping data for analysis
#--Create dataframe with unique invoice number and invoice date
firstPart <- unique(retailonline[,c('InvoiceNo','CustomerID','InvoiceDate')])
#--Create dataframe with unique invoice number and total amount
secondPart <- aggregate(list(Amount=retailonline$Amount),
                        by=list(InvoiceNo=retailonline$InvoiceNo), FUN=sum)
#--merge the two dataframes
dataRFM <- merge(firstPart,secondPart, by="InvoiceNo")
#--reformat InvoiceDate
dataRFM$InvoiceDate <- as.Date(dataRFM$InvoiceDate, format="%m/%d/%Y")
#--Remove rows without Customer ID
dataRFM <- na.omit(dataRFM)
summary(dataRFM)
```

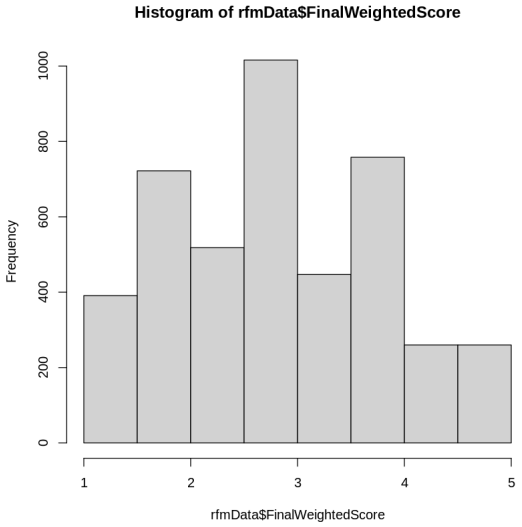
```
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

  InvoiceNo           CustomerID     InvoiceDate            Amount
 Length:22221       Min.   :12346   Min.   :2010-12-01   Min.   :-168469.60
 Class :character   1st Qu.:13755   1st Qu.:2011-03-28   1st Qu.:     88.25
 Mode  :character   Median :15136   Median :2011-07-08   Median :    241.74
                    Mean   :15238   Mean   :2011-06-29   Mean   :    374.39
                    3rd Qu.:16746   3rd Qu.:2011-10-10   3rd Qu.:    418.22
                    Max.   :18287   Max.   :2011-12-09   Max.   : 168469.60
```

```
resultsRFM <- findRFM(dataRFM)
head(resultsRFM[,c('MeanValue','LastTransaction','NoTransaction')])
```

A rowwise_df: 6 × 3

MeanValue	LastTransaction	NoTransaction
<dbl>	<date>	<int>
0.0000	2011-01-18	2
615.7143	2011-12-07	7
449.3100	2011-09-25	4
1757.5500	2011-11-21	1
334.4000	2011-02-02	1
140.4918	2011-11-03	11




```
head(resultsRFM[,c('RecencyScore','FrequencyScore','MonetoryScore',
  'FinalCustomerClass')])
```

A rowwise_df: 6 × 4

RecencyScore	FrequencyScore	MonetoryScore	FinalCustomerClass
<dbl>	<dbl>	<dbl>	<chr>
1	2	1	Class-1
5	4	5	Class-4
2	3	5	Class-3
4	1	5	Class-3
1	1	4	Class-2
3	5	2	Class-3

```
tblClass <- table(resultsRFM$FinalCustomerClass)
tblClass
```

```
Class-1 Class-2 Class-3 Class-4 Class-5
    705    1442    1244     707      84
```

Class 5 = Top 20% of Customers

: There are 84 customers categorized as Class 5

Class 1 = Bottom 20% of Customers

: There are 705 customers categorized as Class 1

```
install.packages('ggplot2')
library(ggplot2)
resultsRFM$Segment <- ifelse(resultsRFM$RecencyScore >= 4 &
                             resultsRFM$FrequencyScore >= 4 &
```

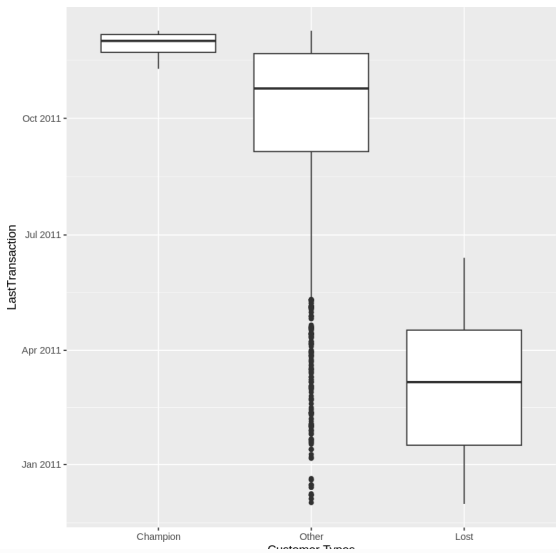
```
                              resultsRFM$MonetoryScore >= 4,
                            "Champion",
                            ifelse(resultsRFM$RecencyScore <= 1 &
                                     resultsRFM$FrequencyScore <= 1,
                                   "Lost",
                                   "Other"))
resultsRFM$Segment <- factor(resultsRFM$Segment, levels = c("Champion", "Other", "Lost"))
summary(resultsRFM$Segment)
```

```
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(as ‘lib’ is unspecified)

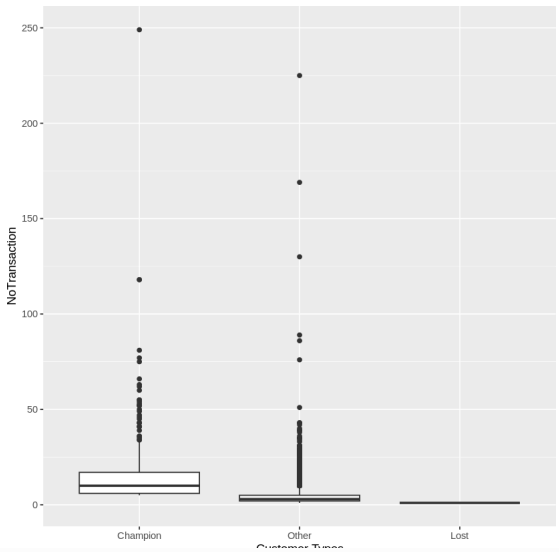
Champion:      412 Other:     3412 Lost:      548
```

Champion customers are those with a recency, frequency, and monetary score all >= 4, whereas lost customers are those with recency and frequency score of 1. There are more lost customers than champion customers.

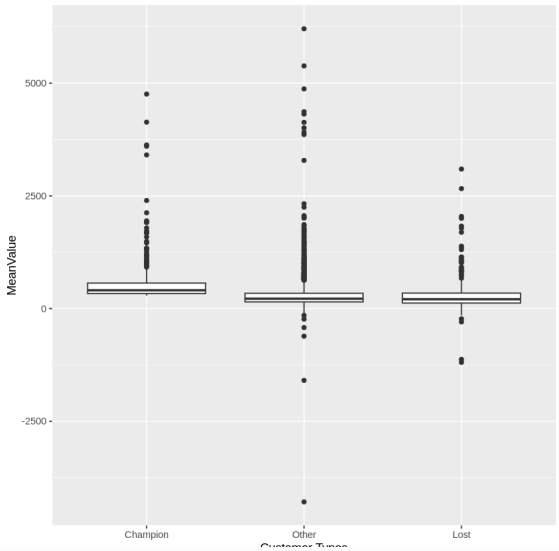
```
#Recency comparison
ggplot(resultsRFM, aes(x=Segment,y=LastTransaction))+geom_boxplot()+labs(x="Customer Types")
```




```
#Frequency comparison
ggplot(resultsRFM,aes(x=Segment,y=NoTransaction))+geom_boxplot()+labs(x="Customer Types")
```




```
#Monetary comparison
ggplot(resultsRFM,aes(x=Segment,y=MeanValue))+geom_boxplot()+labs(x="Customer Types")
```




```
install.packages('dplyr')
library(dplyr)
Champions <- subset(resultsRFM, (RecencyScore>=4 & FrequencyScore>=4 &
                                   MonetoryScore>=4))
Champions$CustomerID <- as.numeric(Champions$CustomerID)
Champions_analysis <- right_join(retailonline, Champions, by='CustomerID')
```

```
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(as ‘lib’ is unspecified)
```

```
Lost <- subset(resultsRFM, (RecencyScore <=1 & FrequencyScore <= 1))
Lost$CustomerID <- as.numeric(Lost$CustomerID)
Lost_analysis <- right_join(retailonline, Lost, by='CustomerID')
```

```
Champions_analysis %>%
group_by(Country) %>%
summarise(totalquantity=sum(Quantity),
                    totalamount=sum(Amount)) %>%
arrange(desc(totalquantity)) %>%
slice_head(n=5)
```

A tibble: 5 × 3

Country	totalquantity	totalamount
<chr>	<dbl>	<dbl>
United Kingdom	1763082	3174486.8
Netherlands	196719	279489.0
EIRE	134205	245956.8
Australia	78704	126506.1
France	62762	113841.0

```
Lost_analysis %>%
group_by(Country) %>%
summarise(totalquantity=sum(Quantity),
                    totalamount=sum(Amount)) %>%
arrange(desc(totalquantity)) %>%
slice_head(n=5)
```

A tibble: 5 × 3

Country	totalquantity	totalamount
<chr>	<dbl>	<dbl>
United Kingdom	76925	129126.60
Ireland	1579	3896.50
Germany	1553	2185.93
Channel Islands	1015	2148.24

We can see that UK, Netherlands, EIRE (Ireland), Australia, and Germany were the Top 5 most ordered countries found among champion customers while UK, France, Greece, Italy, and Germany were ranked as Top 5 among lost customers.

Based on the numbers, there is no doubt that UK is the primary market. Customer service might want to focus on satisfying lost customers in Top 5 countries to see if there are any regional linguistic or cultural issues.

```
Champions_analysis %>%
group_by(Description) %>%
summarise(totalquantity=sum(Quantity),
                  totalamount=sum(Amount)) %>%
arrange(desc(totalamount)) %>%
slice_head(n=5)
```

A tibble: 5 × 3

Description	totalquantity	totalamount
<chr>	<dbl>	<dbl>
REGENCY CAKESTAND 3 TIER	7439	83876.10
WHITE HANGING HEART T-LIGHT HOLDER	19709	54040.05
JUMBO BAG RED RETROSPOT	28634	51892.11
PARTY BUNTING	7195	31485.43
ASSORTED COLOUR BIRD ORNAMENT	19855	30961.03

```
Lost_analysis %>%
group_by(Description) %>%
summarise(totalquantity=sum(Quantity),
                  totalamount=sum(Amount)) %>%
arrange(desc(totalamount)) %>%
slice_head(n=5)
```

A tibble: 5 × 3

Description	totalquantity	totalamount
<chr>	<dbl>	<dbl>
SMALL POPCORN HOLDER	4428	3204.80
REGENCY CAKESTAND 3 TIER	257	3046.35
WHITE HANGING HEART T-LIGHT HOLDER	654	1839.70
RED HARMONICA IN BOX	1569	1695.25
PARTY BUNTING	365	1674.15

REGENCY CAKESTAND 3 TIER, WHITE HANGING HEART T-LIGHT HOLDER and PARTY BUNTING are profitable products regardless of the customer segment, so putting extra attention on how to retain lost customers with strategies such as membership discounts, upselling and cross-selling other popular products would be crucial.